

Application Deployment

(Deploy the given React application to a production ready state)

Application:

Clone the below mentioned repository and deploy the application (Run application in port 3000).

Repo URL : <https://github.com/Vennilavan12/Trend.git>

Docker:

- Dockerize the application by creating Dockerfile
- Build an application and check output using docker image.

Terraform:

- Define infrastructure in main.tf to create VPC, IAM, EC2 with Jenkins, etc.
- Use terraform command to provision infrastructure.

DockerHub:

- Create a DockerHub repository.

Kubernetes:

- Setup Kubernetes in AWS EKS and Confirm EKS cluster is running.
- Write deployment and service YAML files.
- Deploy using kubectl via Jenkins.

Version Control:

- Push the codebase to a Git provider (GitHub).
- Add gitignore and dockerignore files and use CLI commands to push code.

Jenkins:

- Install Jenkins and necessary plugins (Docker, Git, Kubernetes, Pipeline) for build, push & deploy applications.
- Setup Github and jenkins integration using github webhook build trigger for auto build for every commit.
- Create a declarative pipeline script and pipeline project to build, push & deploy using CI-CD.

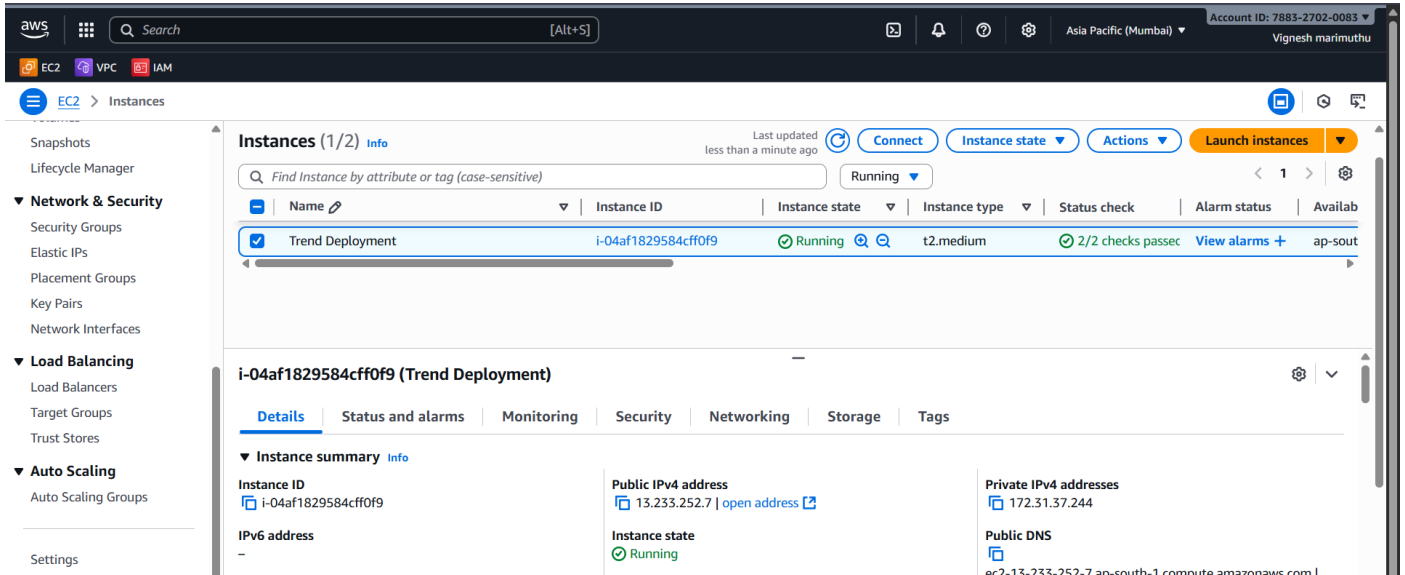
Monitoring:

- Setup a monitoring system to check the health of the cluster or application (opensource) is highly appreciable.

Solution:

✓ 1. Launched an EC2 instance and connected to Ubuntu via SSH

- EC2 will act as the *Jenkins build server and control center* for CI/CD, Docker builds, and EKS deployments.



```
vignesh@vignesh: $ ssh -i "/home/vignesh/.ssh/3_10_key.pem" ubuntu@13.233.252.7
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Oct 10 06:40:16 UTC 2025

System load:  0.0          Processes:      117
Usage of /:   9.4% of 18.33GB Users logged in: 0
Memory usage: 5%          IPv4 address for enx0: 172.31.37.244
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-37-244: $
```

✓ 2. Updated and Installed Required Tools

Installed basic packages

```
ubuntu@ip-172-31-37-244: $ sudo apt-get update -y
sudo apt-get install -y unzip curl wget git
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
```

Installed & Configured AWS CLI

- AWS CLI allows to interact with AWS services (like creating S3 buckets, checking EC2 instances, etc.) directly from the command line.
- It is needed because Terraform, EKS, and Jenkins may use these credentials to communicate with AWS.

```
ubuntu@ip-172-31-37-244: $ curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
unzip awscliv2.zip
sudo ./aws/install
aws --version
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 59.3M 100 59.3M 0 0 123M 0 --:--:-- --:--:-- --:--:-- 123M
Archive: awscliv2.zip
creating: aws/
creating: aws/dist/
inflating: aws/THIRD_PARTY_LICENSES
inflating: aws/README.md
inflating: aws/install
creating: aws/dist/awscli/
creating: aws/dist/dateutil/
creating: aws/dist/docutils/
```

```
You can now run: /usr/local/bin/aws --version
aws-cli/2.31.12 Python/3.13.7 Linux/6.14.0-1011-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-37-244: $ aws configure
AWS Access Key ID [None]: AKIA3PC7TWYZVJBMRPE7
AWS Secret Access Key [None]: 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
Default region name [None]: ap-south-1
Default output format [None]: json
ubuntu@ip-172-31-37-244: $
```

Installed Terraform

- Terraform helps us automate the creation of infrastructure (VPC, Subnet, EC2, Security Groups) with code instead of clicking in AWS Console.
- This is called “Infrastructure as Code (IaC)”.

```
ubuntu@ip-172-31-37-244: $ wget https://releases.hashicorp.com/terraform/1.9.5/terraform_1.9.5_linux_amd64.zip
unzip terraform_1.9.5_linux_amd64.zip
sudo mv terraform /usr/local/bin/
terraform -version
--2025-10-10 06:43:09-- https://releases.hashicorp.com/terraform/1.9.5/terraform_1.9.5_linux_amd64.zip
Resolving releases.hashicorp.com (releases.hashicorp.com)... 108.159.61.93, 108.159.61.121, 108.159.61.8, ...
Connecting to releases.hashicorp.com (releases.hashicorp.com)|108.159.61.93|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27040662 (26M) [application/zip]
Saving to: 'terraform_1.9.5_linux_amd64.zip'

terraform_1.9.5_linux_amd64.zip 100%[=====] 25.79M 138MB/s in 0.2s

2025-10-10 06:43:10 (138 MB/s) - 'terraform_1.9.5_linux_amd64.zip' saved [27040662/27040662]

Archive: terraform_1.9.5_linux_amd64.zip
inflating: LICENSE.txt
inflating: terraform
Terraform v1.9.5
on linux_amd64

Your version of Terraform is out of date! The latest version
is 1.13.3. You can update by downloading from https://www.terraform.io/downloads.html
ubuntu@ip-172-31-37-244: $
```

Installed Docker

- Docker allows us to containerize our React app.
- Container means bundling the app + dependencies together, so it runs the same everywhere.

```
ubuntu@ip-172-31-37-244: $ docker --version
Docker version 28.5.1, build e180ab8
```

Installed kubectl & eksctl

- kubectl is the CLI to interact with Kubernetes clusters — deploy, delete, scale apps.
- eksctl is a helper tool that creates EKS clusters in AWS easily (instead of writing long CloudFormation or Terraform for EKS).

```
ubuntu@ip-172-31-37-244: $ curl -LO "https://dl.k8s.io/release/${curl -Ls https://dl.k8s.io/release/stable.txt}/bin/linux/amd64/kubectl"
chmod +x kubectl
sudo mv kubectl /usr/local/bin/
kubectl version --client
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
100 138    100    138    0     0    434      0 --:--:-- --:--:-- --:--:--    435
100 57.7M 100 57.7M    0     0  74.2M      0 --:--:-- --:--:-- --:--:--  137M
Client Version: v1.34.1
Kustomize Version: v5.7.1
ubuntu@ip-172-31-37-244: $ curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_${uname -s}_amd64.tar.gz" | tar xz
-C /tmp
sudo mv /tmp/eksctl /usr/local/bin
eksctl version
0.215.0
ubuntu@ip-172-31-37-244: $
```

✓ 3. Terraform — Infrastructure Creation

- Instead of manually going to AWS Console to create VPC, Subnets, Security Groups, and EC2, we write Terraform code that can automatically build everything with a single command.

```
ubuntu@ip-172-31-37-244: $ mkdir infra-terraform
cd infra-terraform
ubuntu@ip-172-31-37-244: ~/infra-terraform$
```

```
ubuntu@ip-172-31-37-244: ~/infra-terraform$ cat main.tf
provider "aws" {
  region = "ap-south-1"
}

# Create a VPC (Virtual Private Cloud)
resource "aws_vpc" "main_vpc" {
  cidr_block = "10.0.0.0/16"
}

# Public subnet to host EC2 (Jenkins)
resource "aws_subnet" "public_subnet" {
  vpc_id            = aws_vpc.main_vpc.id
  cidr_block        = "10.0.1.0/24"
  map_public_ip_on_launch = true
  availability_zone  = "ap-south-1a"
}

# Security Group for SSH, HTTP, Jenkins ports
resource "aws_security_group" "jenkins_sg" {
  name        = "jenkins-sg"
  vpc_id      = aws_vpc.main_vpc.id

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    from_port = 8080
    to_port   = 8080
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
```

- I have attached my main.tf file in the GitHub repository:

<https://github.com/vigneshmarimuthu/Trend-v2>

```
ubuntu@ip-172-31-37-244:~/infra-terraform$ terraform init
terraform plan
terraform apply -auto-approve
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.16.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
aws_vpc.main_vpc: Refreshing state... [id=vpc-067f069fd57455533]
aws_subnet.public_subnet: Refreshing state... [id=subnet-0b4bf0e8618aa54bd]
aws_security_group.jenkins_sg: Refreshing state... [id=sg-0ef7fca0ae1cebb45]
```

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

jenkins_public_ip = "13.201.84.124"

```
ubuntu@ip-172-31-37-244:~/infra-terraform$
```

- After applying Terraform using the main.tf file, I got my public IP: 13.201.84.124.

The screenshot shows the AWS Management Console interface. On the left, there's a navigation menu with categories like 'Network & Security', 'Load Balancing', and 'Auto Scaling'. The main area displays 'Instances (1/2)'. A table lists two instances: 'Jenkins-Server' (ID: i-07fd9f5df95f3d163) and 'Trend Deployment' (ID: i-04af1829584cff09), both in 'Running' state. Below the table, the details for the 'Jenkins-Server' instance are shown, including its public IPv4 address (13.201.84.124) and its running status.

✓ 4.Installed the required packages and Jenkins on the EC2 instance (Jenkins-server).

- Jenkins will be the **CI/CD engine** that listens to GitHub commits, builds Docker images, pushes them to DockerHub, and deploys to Kubernetes.

```
PS C:\Users\Vignesh> ssh -i "C:\Users\Vignesh\Downloads\3_10 key.pem" ubuntu@13.201.84.124
The authenticity of host '13.201.84.124 (13.201.84.124)' can't be established.
ED25519 key fingerprint is SHA256:hafMibc6S9ro1xkZWkhplyoQU/cAvztwp+Hd4iZh+8M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.201.84.124' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1016-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Oct 10 07:11:33 UTC 2025

System load:  0.0      Processes:    115
Usage of /:   22.8% of 6.71GB   Users logged in:  0
Memory usage: 5%      IPv4 address for enx0: 10.0.1.143
Swap usage:   0%
```

```

ubuntu@ip-10-0-1-143:~$ sudo systemctl enable jenkins
sudo systemctl start jenkins
sudo systemctl status jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-10-10 07:21:15 UTC; 1min 24s ago
     Main PID: 28366 (java)
        Tasks: 45 (limit: 4676)
      Memory: 515.2M (peak: 611.2M)
         CPU: 20.448s
    CGroup: /system.slice/jenkins.service
            └─28366 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Oct 10 07:21:08 ip-10-0-1-143 jenkins[28366]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Oct 10 07:21:08 ip-10-0-1-143 jenkins[28366]: [LF]>
Oct 10 07:21:08 ip-10-0-1-143 jenkins[28366]: [LF]> *****
Oct 10 07:21:08 ip-10-0-1-143 jenkins[28366]: [LF]> *****
Oct 10 07:21:08 ip-10-0-1-143 jenkins[28366]: [LF]> *****
Oct 10 07:21:15 ip-10-0-1-143 jenkins[28366]: 2025-10-10 07:21:15.123+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Oct 10 07:21:15 ip-10-0-1-143 jenkins[28366]: 2025-10-10 07:21:15.143+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully started
Oct 10 07:21:15 ip-10-0-1-143 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 10 07:21:18 ip-10-0-1-143 jenkins[28366]: 2025-10-10 07:21:18.259+0000 [id=49] INFO h.m.DownloadService$Downloadable#load: Obtained the update
Oct 10 07:21:18 ip-10-0-1-143 jenkins[28366]: 2025-10-10 07:21:18.263+0000 [id=49] INFO hudson.util.Retrier#start: Performed the action check
lines 1-20/20 (END)

```

✓ 5. Accessed Jenkins

✓ 6. Dockerize the React App

- Used NGINX to serve the React build output (dist folder) as static files

Cloned the repo

```

ubuntu@ip-10-0-1-143:~$ git clone https://github.com/Vennilavan12/Trend.git
Cloning into 'Trend'...
remote: Enumerating objects: 77, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 77 (delta 0), reused 0 (delta 0), pack-reused 76 (from 1)
Receiving objects: 100% (77/77), 8.58 MiB | 5.22 MiB/s, done.
Resolving deltas: 100% (1/1), done.

```

Created .dockerignore & .gitignore

- Jenkins will pull from your GitHub repo. .gitignore avoids huge or secret files being committed.
- .dockerignore avoids sending unnecessary files to the Docker daemon during build.

```
ubuntu@ip-10-0-1-143:~/Trend$ nano .gitignore
ubuntu@ip-10-0-1-143:~/Trend$ cat .gitignore
node_modules/
build/
.env
.DS_Store
.vscode/
.idea/
npm-debug.log
ubuntu@ip-10-0-1-143:~/Trend$ nano .dockerignore
ubuntu@ip-10-0-1-143:~/Trend$ cat .dockerignore
node_modules
build
.git
.DS_Store
.env
ubuntu@ip-10-0-1-143:~/Trend$ |
```

Created Dockerfile

```
ubuntu@ip-10-0-1-143:~/Trend$ nano Dockerfile
ubuntu@ip-10-0-1-143:~/Trend$ cat Dockerfile
# Use official Nginx image to serve static files
FROM nginx:alpine

# Remove default Nginx web files
RUN rm -rf /usr/share/nginx/html/*

# Copy your built static files to Nginx's html directory
COPY dist/ /usr/share/nginx/html/

# Expose port 80
EXPOSE 80

# Start Nginx
CMD ["nginx", "-g", "daemon off;"]
ubuntu@ip-10-0-1-143:~/Trend$ |
```

push it to my repo

```
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Enumerating objects: 77, done.
Counting objects: 100% (77/77), done.
Delta compression using up to 2 threads
Compressing objects: 100% (75/75), done.
Writing objects: 100% (77/77), 8.58 MiB | 19.84 MiB/s, done.
Total 77 (delta 1), reused 77 (delta 1), pack-reused 0
remote: Resolving deltas: 100% (1/1), done.
To https://github.com:vigneshmarimuthu/Trend-v2
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
ubuntu@ip-10-0-1-143:~/Trend$ |
```

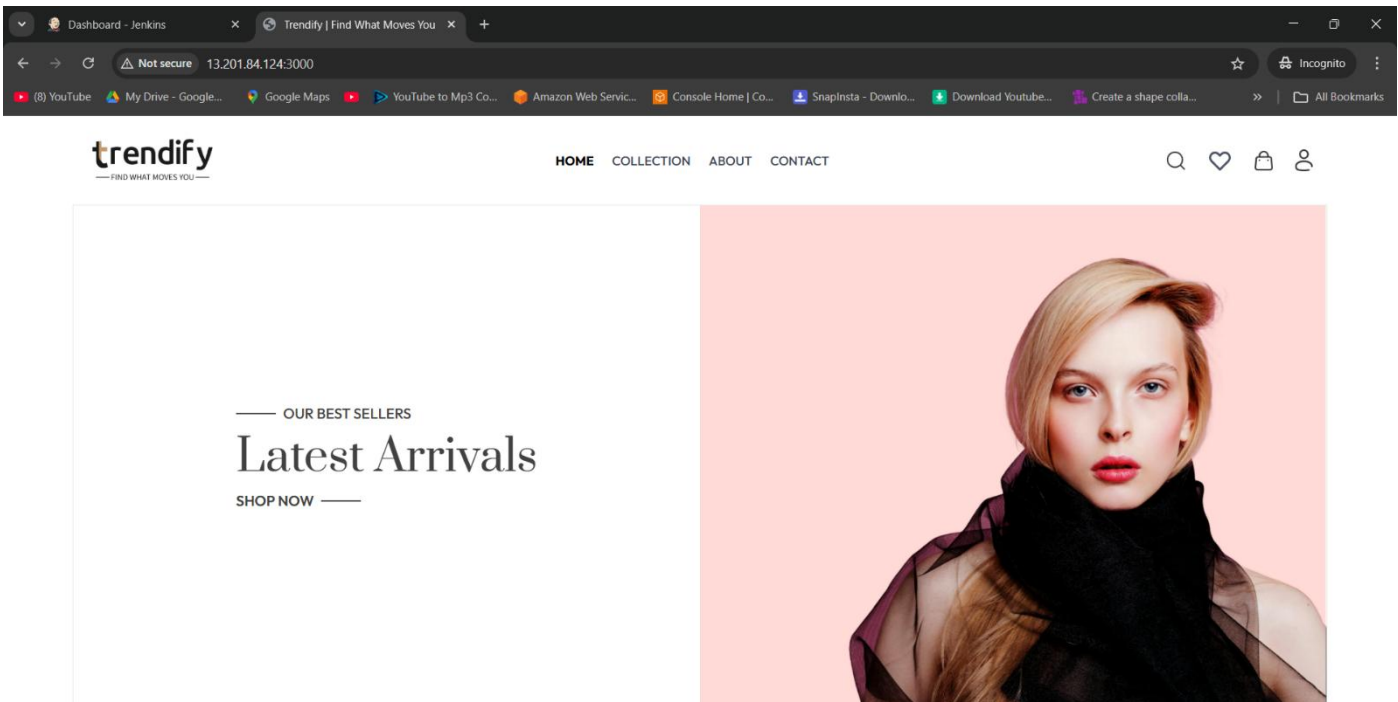
✅ 7. Build & Run

```
ubuntu@ip-10-0-1-143:~/Trend$ docker build -t trend-app .
docker run -d -p 3000:3000 trend-app
[+] Building 5.8s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 221B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [internal] load .dockerignore
=> => transferring context: 79B
=> [internal] load build context
=> => transferring context: 9.24MB
=> [1/3] FROM docker.io/library/nginx:alpine@sha256:7c1b9a91514d1eb5288d7cd6e91d9f451707911bf8a9307a3acbc811d4aa82e
=> resolve docker.io/library/nginx:alpine@sha256:7c1b9a91514d1eb5288d7cd6e91d9f451707911bf8a9307a3acbc811d4aa82e
=> sha256:b03ccb7431a2e3172f5c8ae96d82bd792935f33ecb88fbf2948559e475745c4e 2.50kB / 2.50kB
=> sha256:5e7abcbdd20216bbeedf1369529564ff60f830ed3540c477938ca580b645dfff5 10.78kB / 10.78kB
=> sha256:2d35ebdb57d9971fea0cac1582aa78935adf8858b2cc32db163c98822e5dfa1b 3.80MB / 3.80MB
=> sha256:621a51978ed7f4a65a4324aaf124c21292fd986ed107e731f860572233073798 627B / 627B
=> sha256:7c1b9a91514d1eb5288d7cd6e91d9f451707911bf8a9307a3acbc811d4aa82e 10.33kB / 10.33kB
=> sha256:7c1b9a91514d1eb5288d7cd6e91d9f451707911bf8a9307a3acbc811d4aa82e 10.33kB / 10.33kB
docker:default
0.0s
0.0s
2.2s
0.0s
0.0s
0.1s
0.1s
2.7s
0.0s
0.0s
0.0s
0.8s
0.5s
0.0s
```

```

=> exporting to image
=> => exporting layers
=> => writing image sha256:c2922af45e6e748acf126ec39cecee7568d3b9c84c6e18f73e27756378c8cf2a
=> => naming to docker.io/library/trend-app
0ccfb9c4bf976b1c32245e53c4931aae3e120c49797cc6b169f2902f891c6424
ubuntu@ip-10-0-1-143:~/Trend$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
0ccfb9c4bf97  trend-app "/docker-entrypoint..." 11 seconds ago Up 10 seconds 80/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp magical_borg
ubuntu@ip-10-0-1-143:~/Trend$

```



- It's running on 13.201.84.124:3000

✓ 8. Push Image to DockerHub

- EKS needs to pull the image from a public or private container registry. We'll push it to Docker Hub.

```

ubuntu@ip-10-0-1-143:~/Trend$ docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

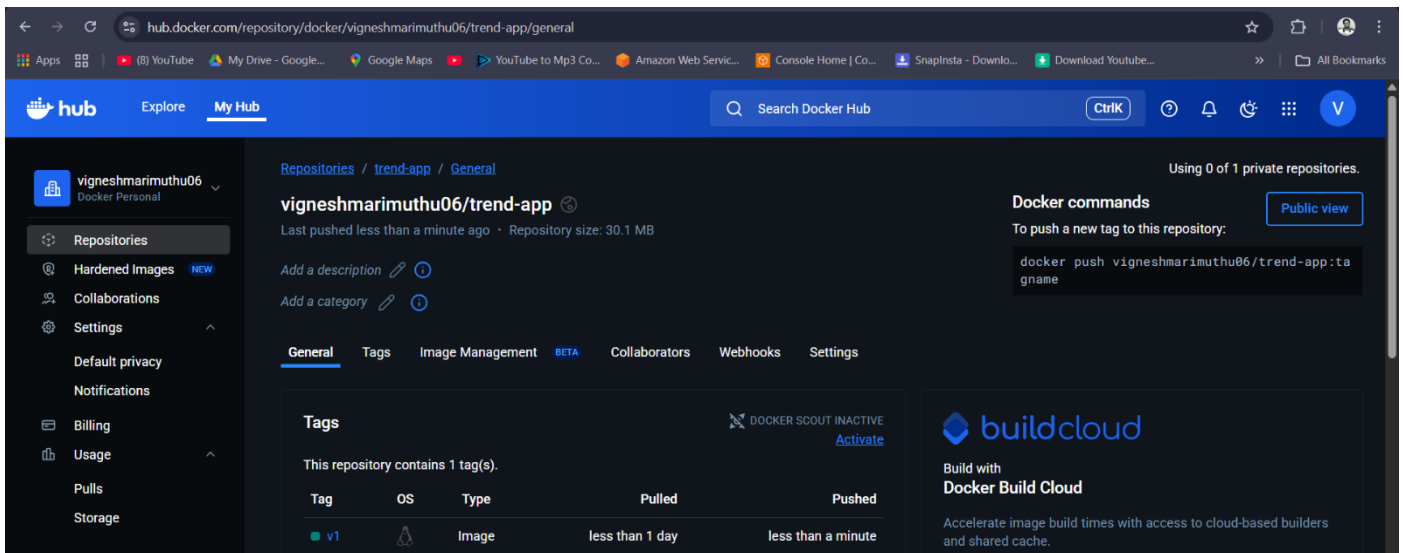
Your one-time device confirmation code is: TSDD-BJQJ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-10-0-1-143:~/Trend$ docker tag trend-app vigneshmarimuthu06/trend-app:v1
ubuntu@ip-10-0-1-143:~/Trend$ docker push vigneshmarimuthu06/trend-app:v1
The push refers to repository [docker.io/vigneshmarimuthu06/trend-app]
9b487eae9147: Pushed
06f582bf7cd7: Pushed
e9a26559275d: Mounted from library/nginx
1704c5a3d8da: Mounted from library/nginx
dc45ec6c902b: Mounted from library/nginx
3e4655a38876: Mounted from library/nginx
abbe7f7f88314: Mounted from library/nginx
ed6be5e2fd33: Mounted from library/nginx
1967fccdbe5e: Mounted from library/nginx
256f393e029f: Mounted from library/nginx
v1: digest: sha256:3307b6f69565e7faf3e56ec0ff2c62b83a7cf9e64da37b10a1f5ad4e62baa335 size: 2407
ubuntu@ip-10-0-1-143:~/Trend$

```



✓ 9. Setup EKS (Kubernetes)

- EKS (Elastic Kubernetes Service) allows us to run and scale containerized applications automatically.

```
ubuntu@ip-10-0-1-143:~/Trend$ eksctl create cluster \
--name trend-cluster \
--region ap-south-1 \
--version 1.28 \
--nodegroup-name standard-workers \
--node-type t3.medium \
--nodes 2
2025-10-10 07:43:07 [i] eksctl version 0.215.0
2025-10-10 07:43:07 [i] using region ap-south-1
2025-10-10 07:43:07 [!] Amazon EKS will no longer publish EKS-optimized Amazon Linux 2 (AL2) AMIs after November 26th, 2025. Additionally, Kubernetes version 1.32 is the last version for which Amazon EKS will release AL2 AMIs. From version 1.33 onwards, Amazon EKS will continue to release AL2023 and Bottlerock based AMIs. The default AMI family when creating clusters and nodegroups in Eksctl will be changed to AL2023 in the future.
2025-10-10 07:43:07 [i] setting availability zones to [ap-south-1a ap-south-1b ap-south-1c]
2025-10-10 07:43:07 [i] subnets for ap-south-1a - public:192.168.0.0/19 private:192.168.96.0/19
2025-10-10 07:43:07 [i] subnets for ap-south-1b - public:192.168.32.0/19 private:192.168.128.0/19
2025-10-10 07:43:07 [i] subnets for ap-south-1c - public:192.168.64.0/19 private:192.168.160.0/19
2025-10-10 07:43:07 [i] nodegroup "standard-workers" will use "" [AmazonLinux2/1.28]
2025-10-10 07:43:07 [!] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2025-10-10 07:43:07 [i] using Kubernetes version 1.28
2025-10-10 07:43:07 [i] creating EKS cluster "trend-cluster" in "ap-south-1" region with managed nodes
2025-10-10 07:43:07 [i] will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2025-10-10 07:43:07 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=ap-south-1 --cluster=trend-cluster'
2025-10-10 07:43:07 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "trend-cluster" in "ap-south-1"
```

```
ubuntu@ip-10-0-1-143:~/Trend$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-51-118.ap-south-1.compute.internal Ready    <none>   3m29s  v1.28.15-eks-113cf36
ip-192-168-85-72.ap-south-1.compute.internal Ready    <none>   3m29s  v1.28.15-eks-113cf36
ubuntu@ip-10-0-1-143:~/Trend$
```

✓ 10. Deploy App to EKS

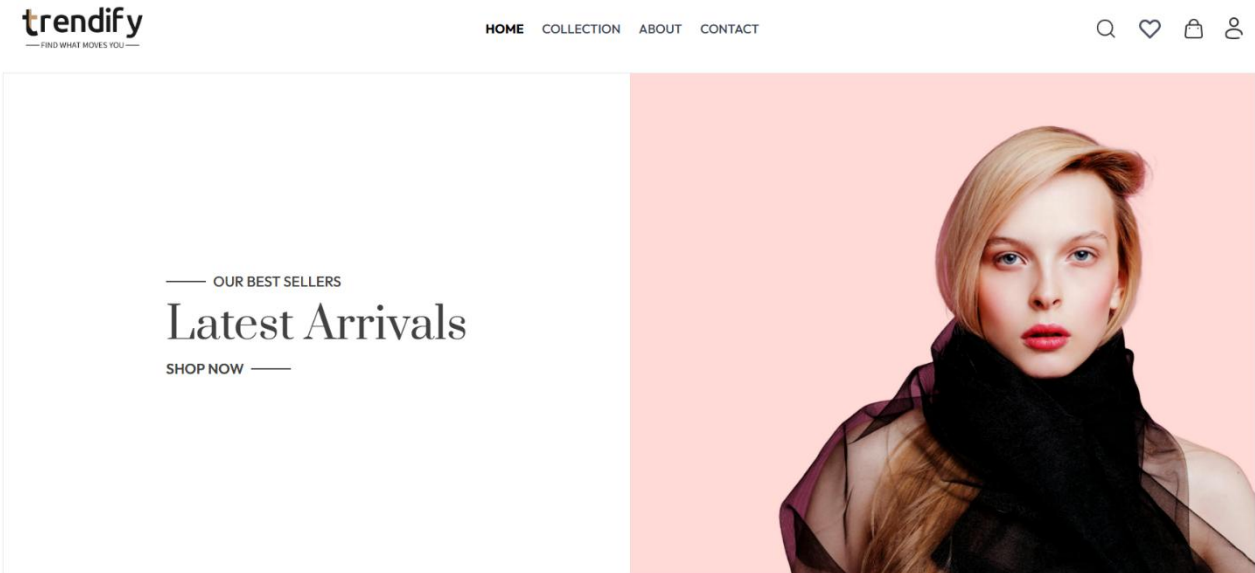
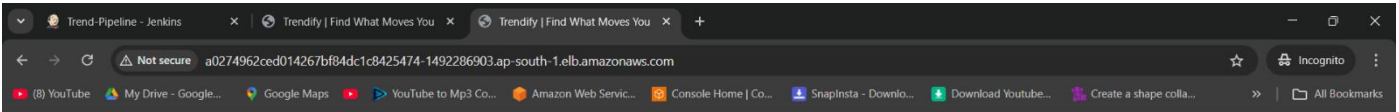
- Deployment → manages pods, replicas, rolling updates.
- Service → exposes app externally.

```
ubuntu@ip-10-0-1-143:~/Trend$ cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: trend-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: trend
  template:
    metadata:
      labels:
        app: trend
    spec:
      containers:
        - name: trend
          image: vigneshmarimuthu06/trend:latest
          ports:
            - containerPort: 3000
ubuntu@ip-10-0-1-143:~/Trend$
```

```
ubuntu@ip-10-0-1-143:~/Trend$ nano service.yml
ubuntu@ip-10-0-1-143:~/Trend$ cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: trend-service
spec:
  type: LoadBalancer
  selector:
    app: trend
  ports:
    - port: 80
      targetPort: 3000
ubuntu@ip-10-0-1-143:~/Trend$
```

```
ubuntu@ip-10-0-1-143:~/Trend$ kubectl apply -f deployment.yml
kubectl apply -f service.yml
deployment.apps/trend-deployment created
service/trend-service created
```

```
^Cubuntu@ip-10-0-1-143:~/Trend$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
trend-deployment-5c6f84bdc-bxl6z    1/1     Running   0           20s
trend-deployment-5c6f84bdc-szlvn    1/1     Running   0           20s
ubuntu@ip-10-0-1-143:~/Trend$ kubectl get svc
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP                                PORT(S)          AGE
kubernetes   ClusterIP   10.100.0.1    <none>                                     443/TCP          19m
trend-service LoadBalancer 10.100.130.186 a0274962ced014267bf84dc1c8425474-1492286903.ap-south-1.elb.amazonaws.com 80:300872/TCP    5m32s
ubuntu@ip-10-0-1-143:~/Trend$
```



11. Jenkins CI/CD Pipeline

- Automate the build → push → deploy steps.
- No more manual builds.

Add credentials for DockerHub

← → ↺ ⚠ Not secure 13.201.84.124:8080/manage/credentials/store/system/domain/_/newCredentials ☆ Incognito

(8) YouTube My Drive - Google... Google Maps YouTube to Mp3 Co... Amazon Web Servic... Console Home | Co... Snapinsta - Downlo... Download Youtube... Create a shape colla... All Bookmarks

Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestricted)

New credentials

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

vigneshmarimuthu06

☐ Treat username as secret ?

Password ?

ID ?

dockerhub-cred

Create

← → ↺ ⚠ Not secure 13.201.84.124:8080/manage/credentials/store/system/domain/_/

(8) YouTube My Drive - Google... Google Maps YouTube to Mp3 Co... Amazon Web Servic... Console Home | Co... Snapinsta - Downlo... Download Youtube... Create a shape colla... All Bookmarks

Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestricted)

Global credentials (unrestricted)

+ Add Credentials

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description
dockerhub-cred	vigneshmarimuthu06/*****	Username with password	

Icon: S M L

Set GitHub webhook

←

→

↺

Not secure

13.201.84.124:8080/job/Trend-Pipeline/configure

☆

Incognito

⋮

(8) YouTube

My Drive - Google...

Google Maps

YouTube to Mp3 Co...

Amazon Web Servic...

Console Home | Co...

SnapInsta - Downlo...

Download Youtube...

Create a shape colla...

»

All Bookmarks

Jenkins

Trend-Pipeline

Configuration

🔍 ⚙️ 👤

Configure

Throttle builds ?

General

Triggers

Pipeline

Advanced

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

☐ Build after other projects are built ?

☐ Build periodically ?

☒ GitHub hook trigger for GITScm polling ?

☐ Poll SCM ?

☐ Trigger builds remotely (e.g., from scripts) ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Save Apply

←

→

↺

Not secure

13.201.84.124:8080/job/Trend-Pipeline/configure

☆

Incognito

⋮

(8) YouTube

My Drive - Google...

Google Maps

YouTube to Mp3 Co...

Amazon Web Servic...

Console Home | Co...

SnapInsta - Downlo...

Download Youtube...

Create a shape colla...

»

All Bookmarks

Jenkins

Trend-Pipeline

Configuration

🔍 ⚙️ 👤

Configure

Pipeline

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/vigneshmarimuthu/Trend-v2

Credentials ?

vigneshmarimuthu06/*****

+ Add

Advanced

Save Apply

Configure

General

Triggers

Pipeline

Advanced

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Add Branch

Repository browser ?

(Auto)

Additional Behaviours

Add

Script Path ?

Jenkinsfile

☒ Lightweight checkout ?

Save

Apply

github.com/vigneshmarimuthu/Trend-v2/settings/hooks/new

vigneshmarimuthu / Trend-v2

Type / to search

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Webhooks / Add webhook

We'll send a POST request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in [our developer documentation](#).

Payload URL *

http://13.201.84.124:8080/github-webhook/

Content type *

application/json

github.com/vigneshmarimuthu/Trend-v2/settings/hooks

vigneshmarimuthu / Trend-v2

Type / to search

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

General

Access

Collaborators

Moderation options

Code and automation

Branches

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

✓ http://13.201.84.124:8080/github-w... (push)

Last delivery was successful.

Edit Delete

Create pipeline job with Jenkinsfile

- Jenkinsfile added to my repo: <https://github.com/vigneshmarimuthu/Trend-v2>.

```
Ubuntu-24.04 x ubuntu@ip-10-0-1-143: ~/Trend x + v
2025-10-10 08:56:58 [I] waiting for CloudFormation stack "eksetl-trend-cluster-nodegroup-standard-workers"
2025-10-10 08:56:58 [I] will delete stack "eksetl-trend-cluster-cluster"
2025-10-10 08:56:58 [✓] all cluster resources were deleted
Command 'terraform' not found, but can be installed with:
sudo snap install terraform
ubuntu@ip-10-0-1-143:~/Trend$ cat Jenkinsfile
pipeline {
    agent any

    environment {
        DOCKERHUB_CREDENTIALS = credentials('dockerhub-cred')
        DOCKERHUB_REPO = "vigneshmarimuthu06/trend"
        EKS_CLUSTER = "trend-cluster"
        AWS_REGION = "ap-south-1"
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/vigneshmarimuthu/Trend-v2.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                sh '''
                    echo "Building Docker image..."
                    docker build -t $DOCKERHUB_REPO:$BUILD_NUMBER .
                '''
            }
        }
    }
}
```

✓ 12. Jenkins build

- Automates code → build → push → deploy. Using BUILD_NUMBER helps track which image came from which Jenkins build.

The screenshot shows the Jenkins web interface for a build named "Trend-Pipeline #3". The interface includes a sidebar with navigation links: Status, Changes, Console Output, Edit Build Information, Delete build '#3', Timings, Git Build Data, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The main content area displays the build status as "Success" with a green checkmark icon. The build number is "#3 (Oct 10, 2025, 8:27:15 AM)". Below the status, it shows the build was started by user "admin" and took 15 seconds to complete. The "This run spent:" section lists the following times: 12 ms waiting, 15 sec build duration, and 15 sec total from scheduled to completion. The "git" section shows the revision "623802a8fa3d805766241172ee60bbb512ccb3b3" and the repository "https://github.com/vigneshmarimuthu/Trend-v2". The "Changes" section shows "No changes." The interface also includes buttons for "Add description" and "Keep this build forever".

Trend-Pipeline #3 Console - Jenkins | Trendify | Find What Moves You | Trendify | Find What Moves You | +

Not secure 13.201.84.124:8080/job/Trend-Pipeline/3/console

Jenkins / Trend-Pipeline / #3 / Console Output

Status
Changes
Console Output
Edit Build Information
Delete build '#3'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build

Console Output

Download Copy View as plain text

```
Started by user admin
Obtained Jenkinsfile from git https://github.com/vigneshmarimuthu/Trend-v2
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/Trend-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
The recommended git tool is: git
using credential dockerhub-cred
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Trend-Pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/vigneshmarimuthu/Trend-v2 # timeout=10
Fetching upstream changes from https://github.com/vigneshmarimuthu/Trend-v2
> git --version # timeout=10
> git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/vigneshmarimuthu/Trend-v2 +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 623802a8fa3d805766241172ee60bbb512ccb3b3 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 623802a8fa3d805766241172ee60bbb512ccb3b3 # timeout=10
Commit message: "configued git identity"
```

Trend-Pipeline #3 Console - Jenkins | Trendify | Find What Moves You | Trendify | Find What Moves You | +

Not secure 13.201.84.124:8080/job/Trend-Pipeline/3/console

Jenkins / Trend-Pipeline / #3 / Console Output

Console Output

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Update kubeconfig & Deploy to EKS)
[Pipeline] sh
+ echo Updating kubeconfig for EKS cluster...
Updating kubeconfig for EKS cluster...
+ aws eks update-kubeconfig --region ap-south-1 --name trend-cluster
Updated context arn:aws:eks:ap-south-1:788327020083:cluster/trend-cluster in /var/lib/jenkins/.kube/config
+ echo Deploying new image to EKS...
Deploying new image to EKS...
+ kubectl set image deployment/trend-deployment trend=vigneshmarimuthu06/trend:3 --record
Flag --record has been deprecated, --record will be removed in the future
deployment.apps/trend-deployment image updated
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Checked if auto triggering works or not

```
ubuntu@ip-10-0-1-143:~/Trend$ echo "Triggered build on $(date)" >> README.md
git add README.md
git commit -m "Trigger Jenkins build"
git push origin main
[main 3200444] Trigger Jenkins build
1 file changed, 1 insertion(+)
create mode 100644 README.md
Username for 'https://github.com': vigneshmarimuthu
Password for 'https://vigneshmarimuthu@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 338 bytes | 338.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/vigneshmarimuthu/Trend-v2.git
623802a..3200444    main -> main
ubuntu@ip-10-0-1-143:~/Trend$
```

Trend-Pipeline #4 Changes - Jenkins

Trendify | Find What Moves You

Trendify | Find What Moves You

Not secure13.201.84.124:8080/job/Trend-Pipeline/4/changes

Jenkins / Trend-Pipeline / #4

Status

Changes

Console Output

Edit Build Information

Delete build '#4'

Polling Log

Timings

Git Build Data

Changes

Summary

1. Trigger Jenkins build (details)

Commit 3200444b13b5d44d2c160868400b98b05d85e550 by vigneshmarimuthu06

Trigger Jenkins build

+ README.md

```
ubuntu@ip-10-0-1-143:~/Trend$ kubectl get svc
NAME          TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes    ClusterIP     10.100.0.1      <none>           443/TCP          45m
trend-service  LoadBalancer 10.100.130.186  a0274962ced014267bf84dc1c8425474-1492286903.ap-south-1.elb.amazonaws.com 80:30872/TCP    31m
ubuntu@ip-10-0-1-143:~/Trend$
```

Trend-Pipeline #5 Console - Jenkins

Trendify | Find What Moves You

Trendify | Find What Moves You

Not securea0274962ced014267bf84dc1c8425474-1492286903.ap-south-1.elb.amazonaws.com

trendify

— FIND WHAT MOVES YOU —

HOME COLLECTION ABOUT CONTACT

Search Heart Shopping Cart User

OUR BEST SELLERS

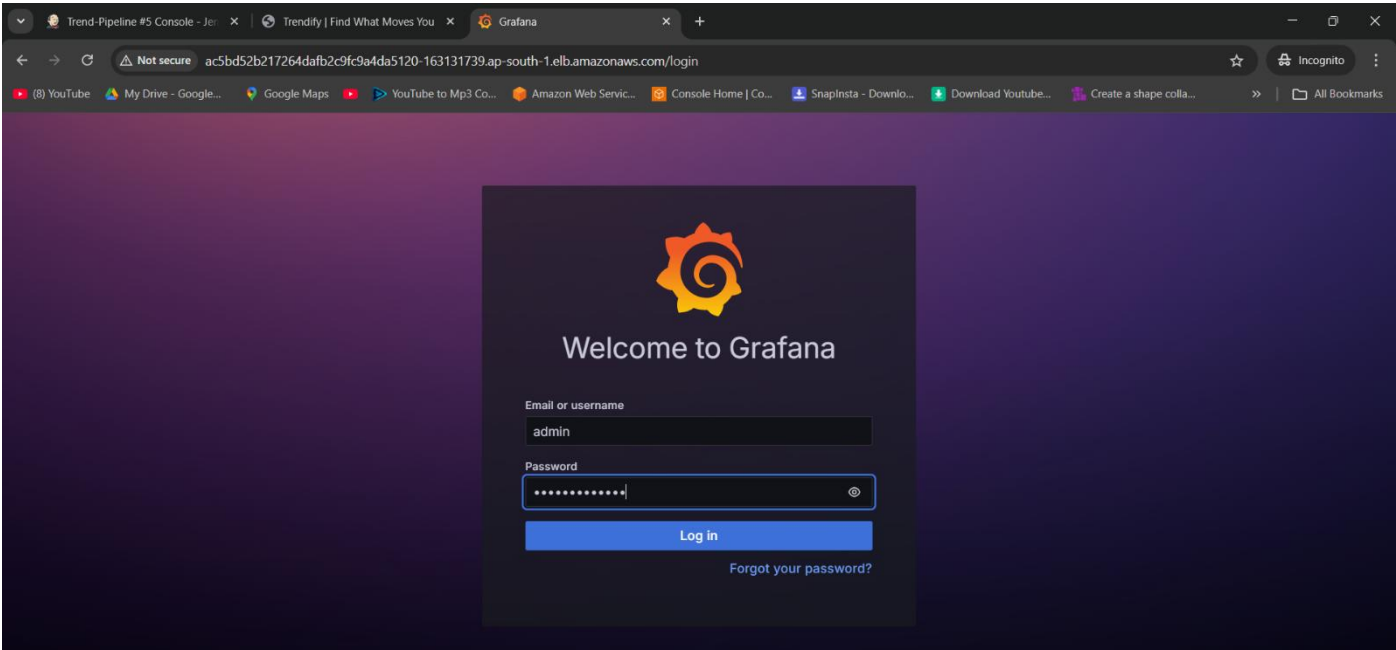
Latest Arrivals

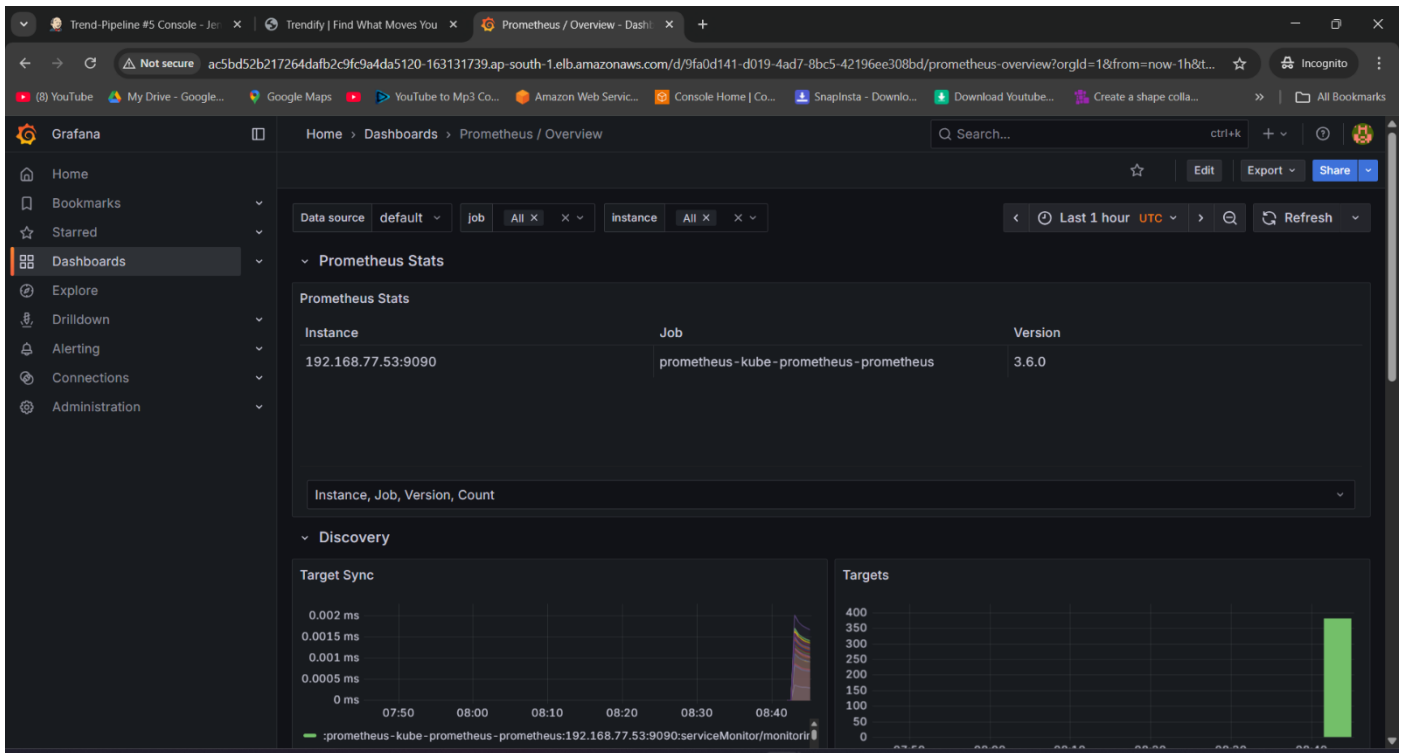
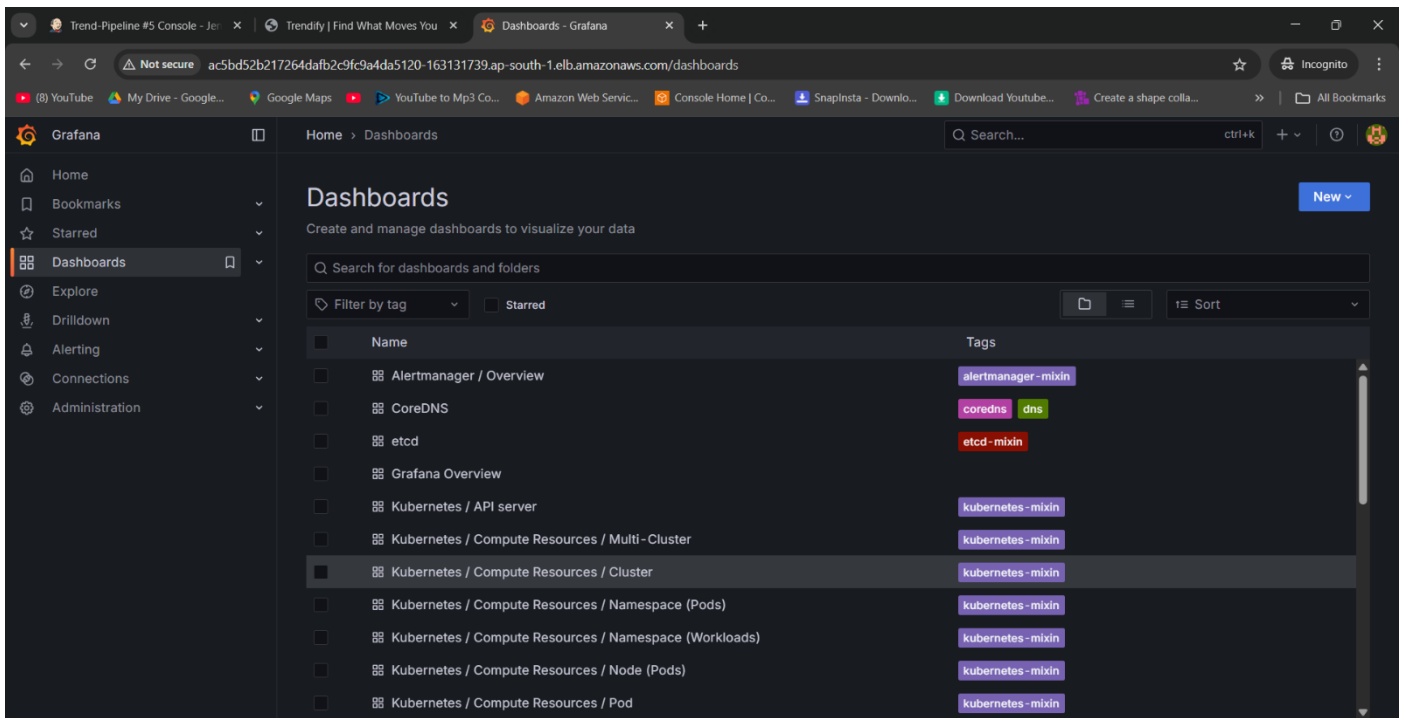
SHOP NOW

✓ 13. Monitoring: Prometheus + Grafana (via Helm)

- To ensure application and cluster are healthy.Prometheus collects metrics; Grafana visualizes them.

```
ubuntu@ip-10-0-1-143:~/Trend$ kubectl get pods -n monitoring -w
NAME                                READY   STATUS    RESTARTS   AGE
alertmanager-prometheus-kube-prometheus-alertmanager-0  2/2     Running   0           38s
prometheus-grafana-77764749c9-ksnsl  3/3     Running   0           43s
prometheus-kube-prometheus-operator-69c5c476b6-xr947    1/1     Running   0           43s
prometheus-kube-state-metrics-7779f5768f-9xntk          1/1     Running   0           43s
prometheus-prometheus-kube-prometheus-prometheus-0     2/2     Running   0           38s
prometheus-prometheus-node-exporter-hfwjc              1/1     Running   0           43s
prometheus-prometheus-node-exporter-q86zp              1/1     Running   0           43s
^Cubuntu@ip-10-0-1-143:~/Trend$ kubectl edit svc prometheus-grafana -n monitoring
service/prometheus-grafana edited
ubuntu@ip-10-0-1-143:~/Trend$ kubectl get svc -n monitoring
NAME                                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)
alertmanager-operated              ClusterIP            10.100.228.156  <none>            9093/TCP
prometheus-grafana                  LoadBalancer        10.100.67.7     ac5bd52b217264dafb2c9fc9a4da5120-163131739.ap-south-1.elb.amazonaws.com  80:30347/TCP
prometheus-kube-prometheus-alertmanager ClusterIP            10.100.228.156  <none>            9093/TCP
prometheus-kube-prometheus-operator ClusterIP            10.100.199.133  <none>            443/TCP
prometheus-kube-prometheus-prometheus ClusterIP            10.100.61.223   <none>            9090/TCP
prometheus-kube-state-metrics      ClusterIP            10.100.43.61    <none>            8080/TCP
prometheus-operated                 ClusterIP            10.100.26.205   <none>            9090/TCP
prometheus-prometheus-node-exporter ClusterIP            10.100.26.205   <none>            9100/TCP
```





- Checked the logs and verified it's working